
Data Assimilation with Stochastic Interpolants

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Abstract

1 Data assimilation combines mathematical models with observational data
2 to estimate the state of complex dynamical systems. However, efficiently
3 sampling from high-dimensional posterior distributions in nonlinear dynam-
4 ical systems remains a major computational challenge. Here we show that
5 stochastic interpolants can be adapted to perform exact posterior sampling
6 for data assimilation without requiring model retraining. Our method de-
7 rives a posterior stochastic differential equation incorporating observational
8 information through drift correction, enabling flexible numerical solvers and
9 tractable likelihood score computation via observation interpolation. Our
10 framework provides superior uncertainty quantification while maintaining
11 computational efficiency. We showcase the efficacy of the approach on series
12 of test cases.

1 Introduction

14 Data assimilation (DA) addresses the fundamental challenge of estimating unknown states
15 or parameters in dynamical systems by optimally combining prior model predictions with
16 noisy observations. This is typically accomplished through either variational methods, which
17 minimize regularized objective functions, or Bayesian methods, which estimate posterior
18 distributions – usually via finite sample approximations Evensen et al. (2022). The Bayesian
19 approach is often preferred due to its natural framework for uncertainty quantification
20 through posterior distribution estimation.

21 Within Bayesian DA, approaches such as ensemble Kalman filters and particle filters are
22 arguably the most popular. Ensemble-based Kalman filters are computationally efficient
23 but require Gaussianity assumptions and can only handle weakly nonlinear forward and
24 observation models. Particle filters, conversely, impose virtually no restrictions but are
25 computationally expensive due to the large number of samples required. This fundamental
26 trade-off between computational tractability and modeling flexibility has motivated extensive
27 research into alternative approaches.

28 One promising direction replaces expensive forward models with efficient neural network
29 surrogates Mücke et al. (2024); Cheng et al. (2022); Zhong et al. (2023); Chen et al. (2023).
30 While achieving substantial computational speed-ups, these deterministic approximations
31 require specifying a priori model error distributions when incorporated into DA frame-
32 works. This poses a fundamental challenge for uncertainty quantification in high-dimensional
33 stochastic systems, where the true model error characteristics are unknown.

34 Recently, generative models have emerged as a compelling alternative, learning to directly
35 sample from distributions of interest. In DA contexts, these models are trained on prior

distributions and subsequently modified to incorporate observational constraints for posterior sampling. Although frameworks like denoising diffusion models Ho et al. (2020) and flow matching models Lipman et al. (2023) have achieved remarkable success in computer vision, their iterative sampling procedures can be computationally demanding, potentially limiting real-time DA applications.

Contributions: We introduce a novel posterior sampling method based on stochastic interpolants (SIs), a recent class of generative models that transform samples between arbitrary distributions via stochastic differential equations. We extend this framework by deriving a posterior SDE that incorporates observational information through drift correction. Our key contributions are:

- We derive a posterior drift expression by incorporating likelihood scores into the tunable diffusion SI framework without requiring retraining.
- We prove that the resulting SDE generates exact posterior samples under ideal conditions.
- We introduce a practical likelihood score estimation method using observation interpolations and derive an optimal diffusion rate that compensates for approximation errors.

2 Related work

2.1 Flow-based generative models in physics applications

Flow-based generative models have received substantial attention in recent years due to their success in generating images and videos. A similar development is occurring in physics applications, where these models are trained to perform probabilistic forecasting by learning to sample from the conditional distribution of the next state given the current state. Several approaches have emerged to address this challenge. Some methods combine deterministic neural operators with noise-based techniques: Lippe et al. (2023) perform initial predictions using neural operators and add/remove noise inspired by denoising diffusion models, while Gao et al. (2024) employ neural networks for time-stepping in latent space followed by physics-informed diffusion upsampling. Other approaches focus on direct conditional sampling. Kohl et al. (2024) trains denoising diffusion models to sample directly from the next state, while Fotiadis et al. (2024) uses flow-matching models for direct conditional sampling. Chen et al. (2024) employs stochastic interpolant models to transform current states into samples from the conditional distribution, which Mücke & Sanderse (2025) extended by incorporating energy consistency into the training.

2.2 Inverse problems with generative models

Generative models are increasingly utilized for solving inverse problems, as detailed in the survey by Daras et al. (2024). For linear inverse problems, several training-free methods have been developed. Pökle et al. (2024) introduces a method for image inverses via flows, while Zhang et al. (2025) utilizes flow priors through iterative corrupted trajectory matching. Recent advancements also include Yan et al. (2024), which proposes Flow with Interpolant Guidance (FIG). In the domain of stochastic interpolation, Ahamed & Haber (2024) developed DAWN-SI for data-aware and noise-informed solutions, and Negrel et al. (2025) explored multitask learning.

2.3 Data assimilation with generative models

The integration of generative models into data assimilation (DA) seeks to overcome the limitations of traditional methods like ensemble Kalman filters and particle filters. Early

81 efforts in this direction include score-based data assimilation by Rozet & Louppe (2023). For
 82 high-dimensional systems, Bao et al. (2024) introduced an ensemble score filter, which was
 83 further extended by Si & Chen (2024) to a latent space for handling sparse observation data.
 84 Multimodal settings have been addressed by Qu et al. (2024) through deep generative DA.
 85 Most relevant to our approach is FlowDAS, introduced by Chen et al. (2025), which utilizes
 86 a stochastic interpolant-based framework for DA.

87 3 Preliminaries

88 We denote discrete physical time steps with superscripts and pseudo-time steps with sub-
 89 scripts. E.g. $\mathbf{u}^n$ denotes a quantity at physical time step n and $\mathbf{x}_\tau$ denotes a quantity
 90 at pseudo-time τ . Furthermore, the notation $\mathbf{u}^{n:m}$, with $n < m$, refers to the time series
 91 $(\mathbf{u}^n, \mathbf{u}^{n+1}, \dots, \mathbf{u}^{m-1}, \mathbf{u}^m)$.

92 3.1 Bayesian data assimilation

93 We briefly outline the Bayesian data assimilation setting in this subsection. For more
 94 information on data assimilation, see Carrassi et al. (2018); Evensen et al. (2022).

95 We consider the discrete dynamical system:

$$\mathbf{u}^n = \mathbf{F}(\mathbf{u}^{n-1}) + \boldsymbol{\xi}^n, \quad \boldsymbol{\xi}^n \sim p_{\boldsymbol{\xi}}, \quad (1a)$$

$$\mathbf{y}^n = \mathbf{H}(\mathbf{u}^n) + \boldsymbol{\eta}^n, \quad \boldsymbol{\eta}^n \sim p_{\boldsymbol{\eta}}, \quad (1b)$$

96 where, for all $n = 1, \dots, N$, $\mathbf{u}^n \in \mathbb{R}^{N_u}$ is the state, $\mathbf{y}^n \in \mathbb{R}^{N_y}$ are the observations, $\boldsymbol{\xi}^n \in \mathbb{R}^{N_u}$
 97 is the model noise, $\boldsymbol{\eta}^n \in \mathbb{R}^{N_y}$ is the observation noise, $\mathbf{F} : \mathbb{R}^{N_u} \rightarrow \mathbb{R}^{N_u}$ is the forward
 98 operator, and $\mathbf{H} : \mathbb{R}^{N_u} \rightarrow \mathbb{R}^{N_y}$ is the observation operator. The system can be the
 99 result of a discretized PDE system and is therefore potentially very high-dimensional. Eq.
 100 (1a) induces a prior transition density, $p(\mathbf{u}^n | \mathbf{u}^{n-1})$, that propagates the state without
 101 incorporating observations. Furthermore, the observation likelihood can be evaluated via the
 102 noise distribution, $p(\mathbf{y}^n | \mathbf{u}^n) = p_{\boldsymbol{\eta}}(\mathbf{y}^n - \mathbf{H}(\mathbf{u}^n))$.

103 In this work, we are interested in the posterior distribution:

$$p(\mathbf{u}^{0:n} | \mathbf{y}^{1:n}) = p(\mathbf{u}^0) \prod_{i=1}^n p(\mathbf{u}^i | \mathbf{y}^i, \mathbf{u}^{i-1}). \quad (2)$$

104 Eq. (2) holds due to the Markovian property of Eq. (1) Carrassi et al. (2018); Evensen et al.
 105 (2022). We can sample from this distribution by autoregressively sampling from the one-step
 106 posterior:

$$p(\mathbf{u}^n | \mathbf{y}^n, \mathbf{u}^{n-1}) = \frac{p(\mathbf{y}^n | \mathbf{u}^n) p(\mathbf{u}^n | \mathbf{u}^{n-1})}{p(\mathbf{y}^n)}. \quad (3)$$

107 Therefore, we focus on sampling from Eq. (3) for the rest of the paper.

108 3.2 Stochastic interpolants for probabilistic forecasting

109 We construct a generative model for probabilistic forecasting using the stochastic interpolant
 110 (SI) framework Albergo et al. (2023); Albergo & Vanden-Eijnden (2023); Chen et al. (2024).
 111 The objective is to generate samples from the forecast prior $p(\mathbf{u}^n | \mathbf{u}^{n-1})$ in Eq. (3) by con-
 112 structing an SDE that transports samples from a base distribution to the target distribution
 113 in pseudo-time $\tau \in [0, 1]$.

114 Consider densities $p(\mathbf{x}_0)$ and $p(\mathbf{x}_1 | \mathbf{x}_0)$ with $\mathbf{x}_0, \mathbf{x}_1 \in \mathbb{R}^d$. The SI framework generates
 115 samples from $p(\mathbf{x}_1 | \mathbf{x}_0)$ by solving the SDE

$$d\mathbf{X}_\tau = \mathbf{b}_\theta(\mathbf{X}_\tau, \mathbf{x}_0, \tau) d\tau + \gamma_\tau d\mathbf{W}_\tau, \quad \tau \in [0, 1], \quad \mathbf{X}_0 = \mathbf{x}_0, \quad (4)$$

116 where $\mathbf{b}_\theta$ is a neural network approximating a drift and γ_τ is a prescribed diffusion coefficient.

117 The central object in the SI framework is the interpolant

$$\mathbf{I}_\tau := \mathbf{I}_\tau(\mathbf{x}_0, \mathbf{x}_1) = \alpha_\tau \mathbf{x}_0 + \beta_\tau \mathbf{x}_1 + \gamma_\tau \mathbf{W}_\tau, \quad (5)$$

118 where $\mathbf{W}_\tau := \sqrt{\tau} \mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$. The scalar functions $\alpha_\tau, \beta_\tau, \gamma_\tau$ satisfy the temporal
 119 boundary conditions $\alpha_0 = \beta_1 = 1$ and $\alpha_1 = \beta_0 = \gamma_1 = 0$, which guarantee that $\mathbf{I}_0 \sim p(\mathbf{x}_0)$
 120 and $\mathbf{I}_1 \sim p(\mathbf{x}_1 | \mathbf{x}_0)$. The stochastic dynamics of the interpolant for given $\mathbf{x}_0$ and $\mathbf{x}_1$ are:

$$\begin{aligned} d\mathbf{I}_\tau &= \underbrace{(\dot{\alpha}_\tau \mathbf{x}_0 + \dot{\beta}_\tau \mathbf{x}_1 + \dot{\gamma}_\tau \mathbf{W}_\tau)}_{=: \mathbf{R}_\tau(\mathbf{x}_0, \mathbf{x}_1)} d\tau + \gamma_\tau d\mathbf{W}_\tau. \end{aligned} \quad (6)$$

121 From the interpolant dynamics one can show that the SDE (4) has the same marginal law
 122 as $\mathbf{I}_\tau$, and thus transports $p(\mathbf{x}_0)$ to $p(\mathbf{x}_1 | \mathbf{x}_0)$, when the drift is given by the conditional
 123 expectation

$$\mathbf{b}(\mathbf{x}, \mathbf{x}_0, \tau) = \mathbb{E}[\mathbf{R}_\tau(\mathbf{x}_0, \mathbf{x}_1) | \mathbf{X}_\tau = \mathbf{x}], \quad (7)$$

124 where the expectation is over $\mathbf{x}_1 \sim p(\mathbf{x}_1 | \mathbf{x}_0)$ and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$. The neural network $\mathbf{b}_\theta$ is
 125 trained to approximate this drift by minimizing the mean-squared-error loss

$$\mathcal{L}(\theta) = \int_0^1 \mathbb{E}[\|\mathbf{b}_\theta(\mathbf{I}_\tau, \mathbf{x}_0, \tau) - \mathbf{R}_\tau\|_2^2] d\tau, \quad (8)$$

126 where the expectation is over $\mathbf{x}_0, \mathbf{x}_1 \sim p(\mathbf{x}_0, \mathbf{x}_1)$, and $\mathbf{z}$, and τ is drawn uniformly from
 127 $[0, 1]$. Crucially, no SDE solve is required during training.

128 Once trained, new samples $\mathbf{X}_1 \sim p(\mathbf{x}_1 | \mathbf{x}_0)$ are generated by numerically solving (4), with
 129 the choice of solver and number of time steps offering a flexible accuracy–cost trade-off.
 130 Setting $\mathbf{x}_0 = \mathbf{u}^{n-1}$ and $\mathbf{x}_1 = \mathbf{u}^n$, the model learns to sample from the one-step forecast
 131 distribution $p(\mathbf{u}^n | \mathbf{u}^{n-1})$, and arbitrarily long trajectories are obtained by iterating this
 132 procedure.

133 4 Methodology

134 In this section we will outline the proposed methodology for posterior sampling with stochastic
 135 interpolants.

136 Let $\mathbf{b}_\theta$ be trained such that solving Eq. (4) with initial condition $\mathbf{x}_0$ and a chosen γ_τ
 137 generates samples from $p(\mathbf{x}_1 | \mathbf{x}_0)$. The aim is then to sample from Eq. (3) by deriving a
 138 modified drift term, $\mathbf{b}_\theta^*$, such that the SDE:

$$\begin{aligned} d\mathbf{X}_\tau^* &= \mathbf{b}_\theta^*(\mathbf{X}_\tau^*, \mathbf{x}_0, \mathbf{y}, \tau) d\tau + \gamma_\tau d\mathbf{W}_\tau, \\ \mathbf{y} &= \mathbf{H}(\mathbf{x}_1) + \boldsymbol{\eta}, \end{aligned} \quad (9)$$

139 with $\tau \in [0, 1]$ and $\mathbf{X}_0 = \mathbf{x}_0$ generates samples from the posterior distribution, $\mathbf{X}_{\tau=1}^* \sim$
 140 $p(\mathbf{x}_1 | \mathbf{y}, \mathbf{x}_0)$. The posterior drift term, $\mathbf{b}_\theta^*$, should ideally not require additional training, but
 141 simply rely on the pre-trained drift term, $\mathbf{b}_\theta$, and the observations, $\mathbf{y}$.

142 4.1 From prior drift to posterior drift

143 To obtain posterior samples rather than prior samples, we add a correction term to the drift:

$$\mathbf{b}_\theta^*(\mathbf{x}, \mathbf{x}_0, \mathbf{y}, \tau) = \mathbf{b}_\theta(\mathbf{x}, \mathbf{x}_0, \tau) + \gamma_\tau^2 \nabla_{\mathbf{x}} \log \underbrace{\mathbb{E}[p(\mathbf{y} | \mathbf{x}_1) | \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0]}_{=\Phi_\tau^{\mathbf{y}}(\mathbf{x}, \mathbf{x}_0)}. \quad (10)$$

144 The added term is the diffusion-scaled likelihood score. It is the expected value of the of the
 145 observation likelihood at the terminal pseudo-time given the current state at pseudo-time
 146 τ . Intuitively, this term pushes the process towards areas with large observation likelihood,

while the prior drift term balances that by pushing the process towards the prior distribution. In total, the process should then produce samples from the posterior distribution.

To show that the stated drift term does indeed generate posterior samples, we have the following theorem:

Theorem 4.1 (Posterior SDE). *Given a drift term, $\mathbf{b}_\theta$, trained using Eq. (8), such that solutions to the SDE in Eq. (4) at time $\tau = 1$, $\mathbf{X}_1$ are distributed according to $p(\mathbf{x}_1 | \mathbf{x}_0)$. Furthermore, given observations, $\mathbf{y}$, likelihood $p(\mathbf{y} | \mathbf{x}_1)$, and function $\Phi_\tau^\mathbf{y}(\mathbf{x}, \mathbf{x}_0) = \mathbb{E}[p(\mathbf{y} | \mathbf{x}_1) | \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0]$, the SDE,*

$$d\mathbf{X}_\tau^* = \mathbf{b}_\theta(\mathbf{X}_\tau^*, \mathbf{x}_0, \tau)d\tau + \gamma_\tau^2 \nabla_{\mathbf{X}_\tau^*} \log \Phi_\tau^\mathbf{y}(\mathbf{X}_\tau^*, \mathbf{x}_0)d\tau + \gamma_\tau d\mathbf{W}_\tau, \quad \tau \in [0, 1], \quad \mathbf{X}_0^* = \mathbf{x}_0, \quad (11)$$

generates solutions with terminal states, $\mathbf{X}_1^$, distributed according to the posterior $p(\mathbf{x}_1 | \mathbf{y}, \mathbf{x}_0)$.*

See Appendix A for a proof.

4.2 Approximating the likelihood score via observation interpolant

To evaluate the posterior drift, we must evaluate the likelihood score at the intermediate SDE states, $\mathbf{x}_\tau$. Evaluating the likelihood at pseudo-time $\tau = 1$ requires computing an integral:

$$\Phi_\tau^\mathbf{y}(\mathbf{x}_\tau, \mathbf{x}_0) = \mathbb{E}[p(\mathbf{y} | \mathbf{x}_1) | \mathbf{X}_\tau = \mathbf{x}_\tau, \mathbf{X}_0 = \mathbf{x}_0] = \int p(\mathbf{y} | \mathbf{x}_1) p(\mathbf{x}_1 | \mathbf{x}_\tau, \mathbf{x}_0) d\mathbf{x}_1 \quad (12)$$

Computing this integral is prohibitively expensive, as it requires sampling from $p(\mathbf{x}_1 | \mathbf{x}_\tau, \mathbf{x}_0)$, which necessitates solving the SDE forward in pseudo-time. Furthermore, we also have to backpropagate through the trajectory in pseudo-time to compute the gradient with respect to $\mathbf{x}_\tau$ to obtain the likelihood score. We propose to overcome this issue by defining an interpolant for the observations, $\bar{\mathbf{y}}_\tau$:

$$\bar{\mathbf{y}}_\tau(\mathbf{x}_0, \mathbf{y}) = \alpha_\tau \mathbf{H}(\mathbf{x}_0) + \beta_\tau \mathbf{y}, \quad (13)$$

where α_τ and β_τ are defined such that:

$$\bar{\mathbf{y}}_0(\mathbf{x}_0, \mathbf{y}) = \mathbf{H}(\mathbf{x}_0), \quad \bar{\mathbf{y}}_1(\mathbf{x}_0, \mathbf{y}) = \mathbf{y}. \quad (14)$$

Then, the expected likelihood at pseudo-time τ can be computed by simply evaluating the likelihood:

$$\bar{\Phi}_\tau^{\bar{\mathbf{y}}_\tau}(\mathbf{x}_\tau, \mathbf{x}_0) = p_\tau(\bar{\mathbf{y}}_\tau | \mathbf{x}_\tau, \mathbf{x}_0) \quad (15)$$

With this formulation, the drift to be used for the posterior SDE is

$$\mathbf{b}_\theta^*(\mathbf{x}_\tau, \mathbf{x}_0, \bar{\mathbf{y}}_\tau, \tau) = \mathbf{b}_\theta(\mathbf{x}_\tau, \mathbf{x}_0, \tau) + \gamma_\tau^2 \nabla_{\mathbf{x}_\tau} \log \bar{\Phi}_\tau^{\bar{\mathbf{y}}_\tau}(\mathbf{x}_\tau, \mathbf{x}_0) \quad (16)$$

Since we are now computing the likelihood with the interpolated observations, we require the distribution of $\bar{\mathbf{y}}_\tau$ given $\mathbf{x}_\tau$:

Lemma 4.2 (Interpolated observation likelihood). *Let the observation operator $\mathbf{H}$ be linear with matrix H , and let the observation noise be Gaussian, $\boldsymbol{\eta} \sim \mathcal{N}(0, R)$. Then the interpolated observation satisfies*

$$\bar{\mathbf{y}}_\tau = H\mathbf{x}_\tau + \bar{\boldsymbol{\eta}}_\tau, \quad \bar{\boldsymbol{\eta}}_\tau := \beta_\tau \boldsymbol{\eta} - \gamma_\tau H\mathbf{W}_\tau, \quad (17)$$

and the conditional distribution, $p(\bar{\mathbf{y}}_\tau | \mathbf{x}_0, \mathbf{x}_\tau)$, can be expressed as a corrected version of the marginal Gaussian $p(\bar{\mathbf{y}}_\tau) = \mathcal{N}(H\mathbf{x}_\tau, \bar{R}_\tau)$, where $\bar{R}_\tau = \beta_\tau^2 R + \gamma_\tau^2 \tau H H^\top$:

$$\mathbb{E}[\bar{\mathbf{y}}_\tau | \mathbf{x}_\tau, \mathbf{x}_0] = H\mathbf{x}_\tau + \mathbf{d}_\tau(\mathbf{x}_\tau, \mathbf{x}_0), \quad \mathbf{d}_\tau(\mathbf{x}_\tau, \mathbf{x}_0) = -\gamma_\tau H \mathbb{E}[\mathbf{W}_\tau | \mathbf{x}_\tau, \mathbf{x}_0], \quad (18)$$

$$\text{Cov}(\bar{\mathbf{y}}_\tau | \mathbf{x}_\tau, \mathbf{x}_0) = \bar{R}_\tau - C_\tau(\mathbf{x}_\tau, \mathbf{x}_0), \quad C_\tau(\mathbf{x}_\tau, \mathbf{x}_0) = \gamma_\tau^2 H (\tau \mathbf{I} - \text{Cov}(\mathbf{W}_\tau | \mathbf{x}_\tau, \mathbf{x}_0)) H^\top. \quad (19)$$

Proof is provided in Appendix B. The conditional expectation of the Wiener process, can be computed directly from the trained drift term.

Lemma 4.3 (Conditional moments of the Wiener process). *The conditional mean and covariance of $\mathbf{W}_\tau$ given $(\mathbf{x}_\tau, \mathbf{x}_0)$ can be expressed in terms of the trained drift $\mathbf{b}_\theta$ as follows:*

$$\mathbb{E}[\mathbf{W}_\tau | \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0] = -\gamma_\tau \tau A_\tau (\beta_\tau \mathbf{b}_\theta(\mathbf{x}, \mathbf{x}_0, \tau) - c_\tau(\mathbf{x}, \mathbf{x}_0)), \quad (20)$$

where $c_\tau(\mathbf{x}, \mathbf{x}_0) = \dot{\beta}_\tau \mathbf{x} + (\beta_\tau \dot{\alpha}_\tau - \dot{\beta}_\tau \alpha_\tau) \mathbf{x}_0$, and $A_\tau = [\tau \gamma_\tau (\dot{\beta}_\tau \gamma_\tau - \beta_\tau \dot{\gamma}_\tau)]^{-1}$. Furthermore,

$$\text{Cov}(\mathbf{W}_\tau | \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0) = \tau \mathbf{I} + \gamma_\tau^2 \tau^2 A_\tau [\beta_\tau J_{\mathbf{b}_\theta}(\mathbf{x}, \mathbf{x}_0, \tau) - \dot{\beta}_\tau \mathbf{I}], \quad (21)$$

where $J_{\mathbf{b}_\theta}(\mathbf{x}, \mathbf{x}_0, \tau) = \nabla_{\mathbf{x}} \mathbf{b}_\theta(\mathbf{x}, \mathbf{x}_0, \tau)$ is the Jacobian of the trained drift with respect to $\mathbf{x}$.

A proof can be found in Appendix B.

Remark 4.4 (Approximate likelihood score). Neglecting the covariance correction, C_τ , yields the computationally convenient approximation:

$$\tilde{p}(\bar{\mathbf{y}}_\tau | \mathbf{x}_\tau) := \mathcal{N}(\bar{\mathbf{y}}_\tau; H\mathbf{x}_\tau + \mathbf{d}_\tau(\mathbf{x}_\tau, \mathbf{x}_0), \bar{R}_\tau), \quad (22)$$

with closed-form likelihood score

$$\nabla_{\mathbf{x}_\tau} \log \tilde{p}(\bar{\mathbf{y}}_\tau | \mathbf{x}_\tau) = H^\top \bar{R}_\tau^{-1} (\bar{\mathbf{y}}_\tau - H\mathbf{x}_\tau - \mathbf{d}_\tau(\mathbf{x}_\tau, \mathbf{x}_0)). \quad (23)$$

This approximation overestimates the covariance (since $C_\tau \succeq 0$) and centers the mean at shifted value $H\mathbf{x}_\tau + \mathbf{d}_\tau$. The total error is absorbed into the approximation bound of Theorem 4.5.

When we use the interpolant likelihood, we commit an approximation error:

Theorem 4.5 (Approximation error). *Let $p_\tau^*(\cdot | \mathbf{y})$ be the marginal distribution of the process, $\mathbf{X}_\tau^*$, satisfying the SDE with the drift using the true likelihood term in Eq. (10), and similarly, let $\bar{p}_\tau(\cdot | \mathbf{y})$ be the marginal distribution of the process, $\bar{\mathbf{X}}_\tau$, satisfying the SDE with the drift using the interpolant likelihood term from Eq. (16). Then the KL divergence between the two distributions is:*

$$D_{\text{KL}}(p_\tau^* || \bar{p}_\tau) = \frac{1}{2} \int_0^1 \gamma_\tau^2 \mathbb{E}_{p_\tau^*} [\|s_\tau(\mathbf{x}_\tau) - \bar{s}_\tau(\mathbf{x}_\tau)\|^2] d\tau, \quad (24)$$

where

$$s_\tau(\mathbf{x}_\tau) = \nabla_{\mathbf{x}_\tau} \log \Phi_\tau^{\mathbf{y}}(\mathbf{x}_\tau, \mathbf{x}_0), \quad \bar{s}_\tau(\mathbf{x}_\tau) = \nabla_{\mathbf{x}_\tau} \log \bar{\Phi}_\tau^{\bar{\mathbf{y}}}(\mathbf{x}_\tau, \mathbf{x}_0). \quad (25)$$

Proof follows directly from Girsanov's theorem.

5 Implementation

6 Results

We consider three test cases. Firstly, a very simple Gaussian test case where an analytical posterior is available. Secondly, a stochastic incompressible Navier Stokes test case. Lastly, a weather forecasting test case with real-world training data. Each test case serve a specific purpose which is outlined in the respective subsections.

6.1 Simple analytical test case

We consider the simple case:

$$\mathbf{u}^1 = \mathbf{u}^0 + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(0, \mathbf{I}), \quad (26a)$$

$$\mathbf{y}^1 = \mathbf{u}^1 + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(0, \mathbf{I}) \quad (26b)$$

Due to the simplicity, we can derive an analytical expression for the posterior. See Appendix C for more details.

In this case, we can derive the true drift term analytically as this is a simplification of the more general case from Proposition B.9 in Chen et al. (2024). Hence, with this setup, we do not need to train any neural networks and can thereby assess the posterior sampling in isolation.

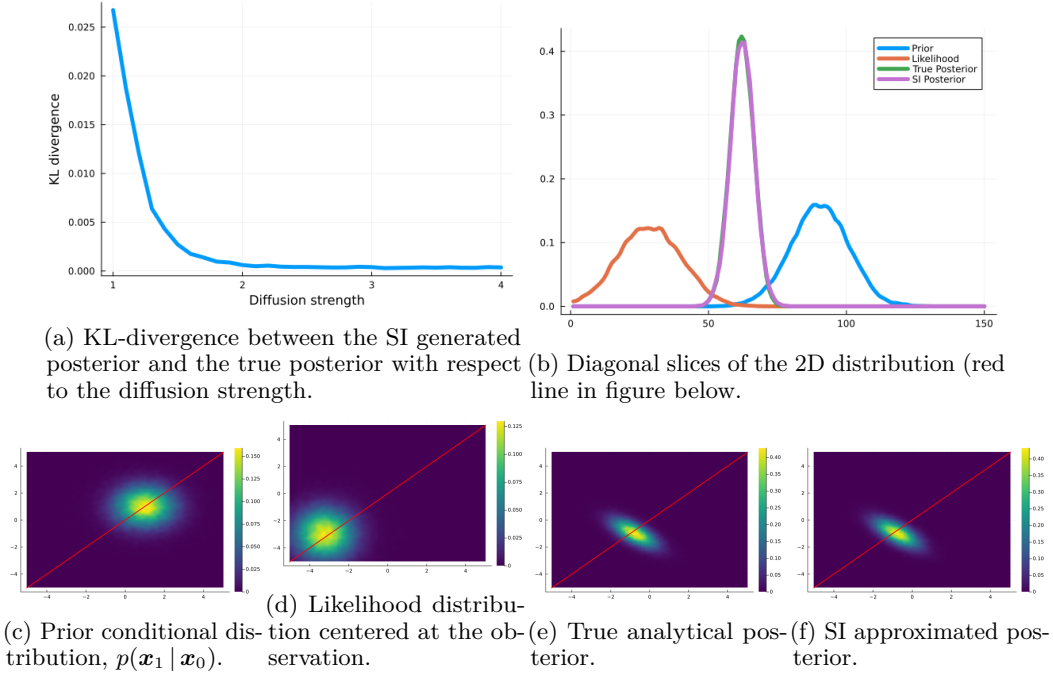


Figure 1: Caption place holder

6.2 Incompressible Navier Stokes

The second test case is an incompressible fluid flow on the torus $\mathcal{T}^2 = [0, 2\pi]^2$, governed by the 2D Navier-Stokes equations with a stochastic forcing term. The problem is formulated in terms of the vorticity field, ω :

$$d\omega + \mathbf{v} \cdot \nabla \omega dt = \nu \Delta \omega dt - \alpha \omega dt + \varepsilon d\xi, \quad (27)$$

The velocity field $\mathbf{v} = \nabla^\perp \psi = (-\partial_y \psi, \partial_x \psi)$ is derived from the stream function $\psi(x, y)$, which satisfies $-\Delta \psi = \omega$. The stochastic term $d\xi$ represents temporally white random forcing applied to selected Fourier modes.

The observation operator is linear:

$$\mathbf{y} = \mathbf{H}(\omega) + \boldsymbol{\eta}, \quad (28)$$

where $\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ represents observation noise with standard deviation $\sigma = 0.05$.

Table 1: Enstrophy, Energy RMSE, and KL-divergence for Navier Stokes test case.

	Vorticity RMSE $32^2 \rightarrow 128^2$	5%	Energy spec RMSE $32^2 \rightarrow 128^2$	5%	KL-Div at points $32^2 \rightarrow 128^2$	5%
FlowDAS						
Guided FM						
Guided diffusion						
OUR METHOD						

6.3 Urban flow

Table 2: Velocity RMSE, temperature RMSE, at KL-divergence at points for the urban test case.

	Vel RMSE		Temp RMSE		KL-Div at points	
	$32^2 \rightarrow 128^2$	5%	$32^2 \rightarrow 128^2$	5%	$32^2 \rightarrow 128^2$	5%
FlowDAS						
Guided FM						
Guided diffusion						
OUR METHOD						

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312 A Proof of Theorem 4.1

313 Before we prove Theorem 4.1, we state Doob’s h -transform Zhang et al.; Särkkä & Solin
 314 (2019):

315 **Proposition A.1** (Doob’s h -transform). *Let $h \in C^2[\mathbb{R}^d]$ be strictly positive, and*

$$\Phi_\tau(\mathbf{x}) = \mathbb{E}[h(\mathbf{X}_T) | \mathbf{X}_\tau = \mathbf{x}], \quad (29)$$

316 *be the solution to the Kolmogorov Backward Equation. Let $p_{\tau,\tau'}(\mathbf{x}, \mathbf{x}')$ be the Markov transition*
 317 *kernel of the SDE:*

$$d\mathbf{X}_\tau = \mathbf{b}(\mathbf{X}_\tau)d\tau + \gamma_\tau d\mathbf{W}_\tau, \quad \tau \in [0, T], \quad \mathbf{X}_0 = \mathbf{x}_0. \quad (30)$$

318 *Furthermore, consider the controlled SDE:*

$$d\mathbf{X}_\tau^u = [\mathbf{b}(\mathbf{X}_\tau^u) + \gamma_\tau \mathbf{u}_\tau(\mathbf{X}_\tau^u)]d\tau + \gamma_\tau d\mathbf{W}_\tau, \quad \tau \in [0, T], \quad \mathbf{X}_0^u = \mathbf{x}_0. \quad (31)$$

319 *Then, if $\mathbf{u}_\tau(\mathbf{x}) = \gamma_\tau \nabla_{\mathbf{x}} \log \Phi_\tau(\mathbf{x})$, the transition kernel is:*

$$p_{\tau,\tau'}^u(\mathbf{x}, \mathbf{x}') = \frac{\Phi_{\tau'}(\mathbf{x}')}{\Phi_\tau(\mathbf{x})} p_{\tau,\tau'}(\mathbf{x}, \mathbf{x}'). \quad (32)$$

320 *In particular, if $\tau = 0$, $\tau' = 1$, and $\mathbf{x} = \mathbf{x}_0$, we have:*

$$p_{0,1}^u(\mathbf{x}, \mathbf{x}') = \frac{h(\mathbf{x}') p_1(\mathbf{x}' | \mathbf{x}_0 = \mathbf{x})}{\Phi_0(\mathbf{x})}, \quad (33)$$

321 *Where $p_1(\mathbf{x}' | \mathbf{x}_0 = \mathbf{x})$ is the marginal density of the process satisfying the uncontrolled SDE*
 322 *at time τ with initial condition $\mathbf{x}_0$.*

323 A.1 Proof of Theorem 4.1

324 *Proof.* Let

$$\mathbf{u}(\mathbf{x}, \tau) = \gamma_\tau \nabla_{\mathbf{x}} \log \Phi_\tau^y(\mathbf{x}, \mathbf{x}_0). \quad (34)$$

325 Then the stochastic interpolant posterior SDE is of the same form as the controlled SDE in
 326 Eq. (31).

327 Assuming perfect training of the drift term, we know that the marginal density, $p_1(\mathbf{x}' | \mathbf{x}_0)$, is
 328 equal to the true prior, $p(\mathbf{x}_1 | \mathbf{x}_0)$.

329 Now, define $h(\mathbf{x}) = p(\mathbf{y} | \mathbf{x})$. Then,

$$\Phi_\tau^y(\mathbf{x}) = \mathbb{E}[p(\mathbf{y} | \mathbf{x}_1) | \mathbf{X}_\tau = \mathbf{x}], \quad (35)$$

330 and,

$$\Phi_0^y(\mathbf{x}_0) = \mathbb{E}[p(\mathbf{y} | \mathbf{x}_1) | \mathbf{X}_0 = \mathbf{x}_0] = \int p(\mathbf{y} | \mathbf{x}_1) p(\mathbf{x}_1 | \mathbf{x}_0) d\mathbf{x}_1, \quad (36)$$

331 Then, via Doob’s h -transform, we have:

$$p_{0,1}^u(\mathbf{x}_0, \mathbf{x}_1) = \frac{p(\mathbf{y} | \mathbf{x}_1) p(\mathbf{x}_1 | \mathbf{x}_0)}{\int p(\mathbf{y} | \mathbf{x}_1) p(\mathbf{x}_1 | \mathbf{x}_0) d\mathbf{x}_1} \quad (37a)$$

$$= \frac{p(\mathbf{y} | \mathbf{x}_1) p(\mathbf{x}_1 | \mathbf{x}_0)}{\int p(\mathbf{y} | \mathbf{x}_1) p(\mathbf{x}_1 | \mathbf{x}_0) d\mathbf{x}_1} \quad (37b)$$

$$= p(\mathbf{x}_1 | \mathbf{y}, \mathbf{x}_0). \quad (37c)$$

332 Which was what we wanted to show.

333 □

334 B Proof of Lemma 4.2 and Lemma 4.3

335 *Proof of Lemma 4.2.* Recall the state interpolant and the observation interpolant:

$$\mathbf{x}_\tau = \alpha_\tau \mathbf{x}_0 + \beta_\tau \mathbf{x}_1 + \gamma_\tau \mathbf{W}_\tau, \quad (38)$$

$$\bar{\mathbf{y}}_\tau = \alpha_\tau H \mathbf{x}_0 + \beta_\tau \mathbf{y}, \quad (39)$$

336 where $\mathbf{W}_\tau = \sqrt{\tau} \mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$, and $\mathbf{y} = H \mathbf{x}_1 + \boldsymbol{\eta}$ with $\boldsymbol{\eta} \sim \mathcal{N}(0, R)$ independent of $\mathbf{z}$.

337 Substituting $\mathbf{y} = H \mathbf{x}_1 + \boldsymbol{\eta}$ into (39):

$$\bar{\mathbf{y}}_\tau = H(\alpha_\tau \mathbf{x}_0 + \beta_\tau \mathbf{x}_1) + \beta_\tau \boldsymbol{\eta}. \quad (40)$$

338 From (38), $\alpha_\tau \mathbf{x}_0 + \beta_\tau \mathbf{x}_1 = \mathbf{x}_\tau - \gamma_\tau \mathbf{W}_\tau$, giving

$$\bar{\mathbf{y}}_\tau = H \mathbf{x}_\tau + \underbrace{\beta_\tau \boldsymbol{\eta} - \gamma_\tau H \mathbf{W}_\tau}_{=\bar{\boldsymbol{\eta}}_\tau}. \quad (41)$$

339 Since $\boldsymbol{\eta}$ is independent of both $\mathbf{x}_1$ and $\mathbf{z}$, we have $\boldsymbol{\eta} \perp (\mathbf{W}_\tau, \mathbf{x}_\tau) \mid \mathbf{x}_0$. Therefore $\boldsymbol{\eta} \perp \mathbf{x}_\tau \mid \mathbf{x}_0$,
340 and

$$\mathbb{E}[\boldsymbol{\eta} \mid \mathbf{x}_\tau, \mathbf{x}_0] = 0, \quad \text{Cov}(\boldsymbol{\eta} \mid \mathbf{x}_\tau, \mathbf{x}_0) = R. \quad (42)$$

341 Using (41) and the independence of $\boldsymbol{\eta}$:

$$\begin{aligned} \mathbb{E}[\bar{\mathbf{y}}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0] &= H \mathbf{x}_\tau + \beta_\tau \mathbb{E}[\boldsymbol{\eta} \mid \mathbf{x}_\tau, \mathbf{x}_0] - \gamma_\tau H \mathbb{E}[\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0] \\ &= H \mathbf{x}_\tau - \gamma_\tau H \mathbb{E}[\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0]. \end{aligned} \quad (43)$$

342 The bias term is $\mathbf{d}_\tau = -\gamma_\tau H \mathbb{E}[\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0]$, which is nonzero whenever $\gamma_\tau > 0$ and
343 $\mathbb{E}[\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0] \neq 0$.

344 Since $\boldsymbol{\eta} \perp (\mathbf{W}_\tau, \mathbf{x}_\tau) \mid \mathbf{x}_0$, the cross-terms between $\boldsymbol{\eta}$ and $\mathbf{W}_\tau$ vanish upon conditioning on
345 $(\mathbf{x}_\tau, \mathbf{x}_0)$:

$$\begin{aligned} \text{Cov}(\bar{\boldsymbol{\eta}}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0) &= \beta_\tau^2 \text{Cov}(\boldsymbol{\eta} \mid \mathbf{x}_\tau, \mathbf{x}_0) + \gamma_\tau^2 H \text{Cov}(\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0) H^\top \\ &= \beta_\tau^2 R + \gamma_\tau^2 H \text{Cov}(\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0) H^\top. \end{aligned} \quad (44)$$

346 Since the marginal covariance is $\text{Cov}(\mathbf{W}_\tau) = \tau \mathbf{I}$, the correction is

$$C_\tau = \bar{R}_\tau - \text{Cov}(\bar{\boldsymbol{\eta}}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0) = \gamma_\tau^2 H (\tau \mathbf{I} - \text{Cov}(\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0)) H^\top. \quad (45)$$

347 By the law of total variance, $\text{Cov}(\mathbf{W}_\tau) = \mathbb{E}[\text{Cov}(\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0)] + \text{Cov}(\mathbb{E}[\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0])$, which
348 implies $\tau \mathbf{I} - \text{Cov}(\mathbf{W}_\tau \mid \mathbf{x}_\tau, \mathbf{x}_0) \succeq 0$ in expectation, so $C_\tau \succeq 0$ on average, confirming that
349 the surrogate covariance $\bar{R}_\tau$ overestimates the true conditional covariance. $\square$

350 *Proof of Lemma 4.3.* We write $\mathbf{W}_\tau = \sqrt{\tau} \mathbf{z}$ and work with the standard normal $\mathbf{z}$. All
351 expectations below are conditioned on $(\mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0)$.

352 **Conditional mean.** Via Stein's formula Albergo et al. (2023); Chen et al. (2024), the
353 conditional expectation of $\mathbf{z}$ relates to the score:

$$\mathbb{E}[\mathbf{z} \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0] = -\gamma_\tau \sqrt{\tau} \nabla_{\mathbf{x}} \log p_\tau(\mathbf{x} \mid \mathbf{x}_0). \quad (46)$$

354 Therefore,

$$\mathbb{E}[\mathbf{W}_\tau \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0] = \sqrt{\tau} \mathbb{E}[\mathbf{z} \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0] = -\gamma_\tau \tau \nabla_{\mathbf{x}} \log p_\tau(\mathbf{x} \mid \mathbf{x}_0). \quad (47)$$

355 Using the relation from Chen et al. (2024),

$$\nabla_{\mathbf{x}} \log p_\tau(\mathbf{x} \mid \mathbf{x}_0) = A_\tau [\beta_\tau \mathbf{b}_\theta(\mathbf{x}, \mathbf{x}_0, \tau) - c_\tau(\mathbf{x}, \mathbf{x}_0)], \quad (48)$$

356 and substituting into (47), we obtain:

$$\mathbb{E}[\mathbf{W}_\tau \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0] = -\gamma_\tau \tau A_\tau (\beta_\tau \mathbf{b}_\theta(\mathbf{x}, \mathbf{x}_0, \tau) - c_\tau(\mathbf{x}, \mathbf{x}_0)). \quad (49)$$

357 **Conditional covariance.** The interpolant can be written as $\mathbf{x}_\tau = \mathbf{m}_\tau + \sigma_\tau \mathbf{z}$, where
 358 $\mathbf{m}_\tau = \alpha_\tau \mathbf{x}_0 + \beta_\tau \mathbf{x}_1$ and $\sigma_\tau = \gamma_\tau \sqrt{\tau}$. For fixed $\mathbf{x}_0$, the marginal density of $\mathbf{x}_\tau$ is a Gaussian
 359 mixture (over $\mathbf{x}_1$) with per-component variance $\sigma_\tau^2 \mathbf{I}$. The second-order Tweedie identity
 360 gives the conditional covariance of $\mathbf{m}_\tau$:

$$\text{Cov}(\mathbf{m}_\tau \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0) = \sigma_\tau^2 \mathbf{I} + \sigma_\tau^4 \nabla_{\mathbf{x}}^2 \log p_\tau(\mathbf{x} \mid \mathbf{x}_0). \quad (50)$$

361 Since $\mathbf{z} = (\mathbf{x}_\tau - \mathbf{m}_\tau)/\sigma_\tau$, we have:

$$\text{Cov}(\mathbf{z} \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0) = \frac{1}{\sigma_\tau^2} \text{Cov}(\mathbf{m}_\tau \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0) \quad (51)$$

$$= \mathbf{I} + \sigma_\tau^2 \nabla_{\mathbf{x}}^2 \log p_\tau(\mathbf{x} \mid \mathbf{x}_0). \quad (52)$$

362 Then, since $\mathbf{W}_\tau = \sqrt{\tau} \mathbf{z}$:

$$\text{Cov}(\mathbf{W}_\tau \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0) = \tau (\mathbf{I} + \gamma_\tau^2 \nabla_{\mathbf{x}}^2 \log p_\tau(\mathbf{x} \mid \mathbf{x}_0)). \quad (53)$$

363 Taking the Jacobian of the score with respect to $\mathbf{x}$:

$$\nabla_{\mathbf{x}}^2 \log p_\tau(\mathbf{x} \mid \mathbf{x}_0) = A_\tau [\beta_\tau J_{\mathbf{b}_\theta}(\mathbf{x}, \mathbf{x}_0, \tau) - \dot{\beta}_\tau \mathbf{I}], \quad (54)$$

364 where we used $\nabla_{\mathbf{x}} c_\tau(\mathbf{x}, \mathbf{x}_0) = \dot{\beta}_\tau \mathbf{I}$. Substituting into (53) yields:

$$\text{Cov}(\mathbf{W}_\tau \mid \mathbf{X}_\tau = \mathbf{x}, \mathbf{X}_0 = \mathbf{x}_0) = \tau \mathbf{I} + \gamma_\tau^2 \tau^2 A_\tau [\beta_\tau J_{\mathbf{b}_\theta}(\mathbf{x}, \mathbf{x}_0, \tau) - \dot{\beta}_\tau \mathbf{I}]. \quad (55)$$

365 □

366 C Simple test case

367 The analytical drift term for the SI SDE that transforms $\mathbf{x}_0$ samples to $\mathbf{x}_1$ samples is:

$$\mathbf{b}(\mathbf{x}, \mathbf{x}_0, \tau) = \dot{\alpha}_\tau \mathbf{x}_0 + \dot{\beta}_\tau \bar{\boldsymbol{\mu}}(\mathbf{x}_0) + (\beta_\tau \dot{\beta}_\tau \mathbf{I} + \tau \gamma_\tau \dot{\gamma}_\tau \mathbf{I}) \bar{\boldsymbol{\Sigma}}^{-1}(\mathbf{x} - \bar{\boldsymbol{\mu}}(\mathbf{x}_0)), \quad (56)$$

368 where

$$\bar{\boldsymbol{\mu}}(\mathbf{x}_0) = \alpha_\tau \mathbf{x}_0 + \beta_\tau \mathbf{x}_0, \quad \bar{\boldsymbol{\Sigma}} = \beta_\tau^2 \mathbf{I} + \tau \gamma_\tau^2 \mathbf{I}. \quad (57)$$

369 The true posterior is:

$$p(\mathbf{u}^1 \mid \mathbf{y}^1) = \mathcal{N}(\mathbf{u}^0 + K(\mathbf{y}^1 - \mathbf{u}^0), \mathbf{I} - K), \quad (58)$$

370 where

$$K = H^\top (H H^\top + \mathbf{I})^{-1} = 0.5 \mathbf{I}. \quad (59)$$